

Naveen Khan

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Professional Summary

Computer Systems Engineering graduate specializing in Artificial Intelligence, Machine Learning, and AI Automation. Skilled in Python, large language model integration, and RAG-based retrieval systems, with hands-on experience building intelligent solutions for real-world organizational needs. Committed to delivering practical, scalable AI products that lower operational costs and drive measurable impact.

Education

Mehran University of Engineering & Technology

Bachelor of Engineering in Computer Systems Engineering | CGPA: 3.6/4.0

Jamshoro, Pakistan

Nov 2021 – Jan 2026

Work Experience

AI Engineer Intern

Civil Aviation Authority of Pakistan (CAA)

Jul 2025 – Aug 2025

Hyderabad, Pakistan (On-site)

- Built an enterprise conversational system using **LLMs** and **RAG**, reducing query resolution time by **60%** and cutting operational support costs by **30%**.
- Designed a semantic retrieval framework with embedding-based document indexing across **50+ organizational PDFs**, improving retrieval precision by **35%**.
- Optimized prompt configurations and evaluation benchmarks, reducing unnecessary API calls by **25%** and lowering inference costs.

AI Engineer Intern

Itsolera Pvt. Ltd.

Jan 2026 – Apr 2026

Islamabad, Pakistan (Remote)

- Developed and fine-tuned **DenseNet121 CNN** models for medical image classification, attaining **93% evaluation accuracy** through systematic hyperparameter experimentation.
- Oversaw full-cycle ML workflows – dataset augmentation, training, cross-validation, and deployment – ensuring production-ready healthcare solutions.
- Shipped a clinician-friendly diagnostic tool that generates automatic predictions with confidence scores.

Projects

AI Chatbot Assistant (CATI)

Retrieval Augmented Generation, FastAPI, NLP, LangChain

- Architected a domain-adaptive dialogue agent integrating **Mistral LLM**, **Groq API**, and **Hugging Face** embeddings with contextual RAG retrieval workflows.
- Built an efficient **FAISS** semantic search system for **50+ documents**, cutting embedding and infrastructure costs while improving response efficiency by **40%**.

Machdollas AI Customer Support Agent

n8n, Google Gemini, Webhooks, HTML/CSS/JS, Vercel

- Launched a 24/7 autonomous AI agent using **n8n** and **Google Gemini**, reducing manual customer service workload by **65%** while handling customer queries and order management.
- Connected **Google Sheets** as a live backend for menu retrieval and order tracking, lowering order error rate by **40%** and delivering a responsive chat interface on **Vercel**.

Medical Disease Diagnosis Agent

TensorFlow, Keras, CNNs, Streamlit

- Trained a **DenseNet121** classifier on MRI data, achieving **95% training** and **93% validation accuracy** with zero overfitting via augmentation and regularization.
- Shipped a clinician-facing **Streamlit** interface for instant MRI/X-ray diagnostic predictions with confidence scores.

Multimodal Smart Wearable for Personal Safety

ESP32, Raspberry Pi 4, IoT, OpenCV, GSM/GPS

- Implemented a **YOLO**-based vision detection system reaching **91% precision** in threat recognition, with multilingual voice commands and a panic button for rapid emergency response.
- Added geospatial tracking and automated alert mechanisms; **awarded 2nd place** at IEEE CS Society Project Exhibition among 45+ projects, MUET 2025.

Technical Skills

Programming & AI: Python, Machine Learning, RAG Pipeline, LLM APIs, Computer Vision, Data Augmentation, Data Visualization, Feature Engineering, EDA, Performance Tuning, Evaluation Metrics

AI Automation: n8n Workflow Automation, AI Agents, API Integration, Webhooks, Large Language Model (LLM) Integrations (OpenAI, Gemini, Claude), Embeddings, RAG System Integration

Data Infrastructure: Relational Databases, MySQL, SQL Workbench, Vector Databases (Pinecone, FAISS)

Tools: Streamlit, Roboflow, Git/GitHub, Google Colab, Jupyter Notebook, VS Code, Vercel